

Maintenance Wizard Report

Diagnosis

The CCM Segment Bearing is in an advanced state of failure, indicated by high vibration, increasing temperature, and abnormal noise. This strongly suggests internal damage, wear, or lubrication breakdown.

Root Cause

Probable root causes include fatigue wear from prolonged operation, inadequate or contaminated lubrication, misalignment, excessive load, or ingress of foreign particles. A detailed failure analysis post-replacement will be necessary to pinpoint the exact cause.

Risk Level

Critical

Failure Probability

Very High (imminent failure)

Remaining Useful Life

Hours to a few days at best; catastrophic failure could occur at any moment under load.

Catastrophic Risk

High. Potential for sudden bearing seizure, leading to segment deformation or cracking, roller damage, strand collapse, unplanned emergency shutdown of the entire caster, extensive production loss, and significant safety hazards due to ejected components or sudden machine movement.

Immediate Actions

['Initiate an emergency shutdown procedure for the affected strand/machine if symptoms rapidly escalate or breach critical alarm limits.', 'Isolate power and apply Lockout/Tagout (LOTO) procedures.', 'Conduct a rapid visual inspection for external signs of damage, lubricant leakage, or excessive heat on the bearing housing.', 'Notify operations and maintenance management immediately regarding the critical status.', 'Prepare all necessary resources (personnel, tools, lifting equipment, spares) for segment removal and replacement.']

Short Term Actions

['Schedule an urgent, planned outage for the affected strand to replace the faulty CCM segment bearing and/or the entire segment.', 'Verify the availability and integrity of the spare CCM segment bearing and any associated components (seals, fasteners).', 'Perform a thorough inspection of the removed segment and surrounding components for any secondary damage (e.g., roller wear,

housing distortion, shaft damage) that might have resulted from the bearing failure.']

Long Term Actions

['Conduct a comprehensive Root Cause Analysis (RCA) on the failed bearing to identify underlying factors (e.g., lubrication schedule/quality, operating conditions, material defects, installation practices, environmental contamination).', 'Review and optimize the lubrication program, including lubricant type, application frequency, and method, for all CCM segment bearings.', 'Implement or enhance condition monitoring techniques (e.g., continuous vibration analysis, thermal imaging, acoustic monitoring) for critical bearings in the CCM.', 'Evaluate current bearing specifications and consider alternative designs or materials if failures are recurrent.', 'Provide refresher training to maintenance personnel on proper bearing installation, removal, lubrication, and proactive inspection techniques.']

Spare Procurement

['CCM Segment Bearing (ensure critical spares are always on hand)', 'Associated seals, spacers, and retention hardware.', 'High-temperature, heavy-duty lubricant compatible with CCM operating conditions.', 'Specialized bearing installation and removal tools (e.g., induction heater, hydraulic puller, alignment tools).']

Executive Summary

A CCM Segment Bearing is exhibiting severe symptoms of impending failure, including high vibration, increasing temperature, and abnormal noise. This presents a critical risk of imminent catastrophic failure, which could lead to significant production losses, extensive equipment damage, and serious safety hazards. Immediate intervention is imperative to mitigate these risks.

Maintenance Recommendation

It is strongly recommended to initiate an immediate, planned shutdown of the affected strand/machine to replace the failing CCM Segment Bearing. Following replacement, a comprehensive root cause analysis must be conducted, and long-term preventive measures, including enhanced condition monitoring and optimized lubrication, should be implemented across all CCM segments to prevent recurrence.

Conclusion

The current state of the CCM Segment Bearing indicates an extremely high probability of imminent and catastrophic failure. Prompt and decisive action is critical to prevent severe operational disruptions, costly secondary damage, and potential safety incidents. Failure to address this immediately will result in significantly higher costs and risks.